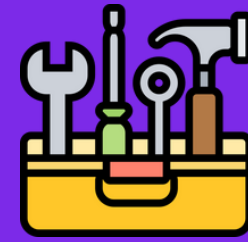


RAG

Retrieval-Augmented Generation



1. Preparation – done once

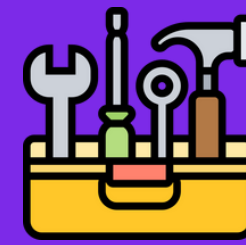


- Split document into small chunks
- Convert each chunk → embedding (numerical vector)
- Save embeddings in vector store (fast semantic search)

Keyword search vs Semantic search

- **Keyword** = exact words match → misses synonyms ("digital marketing" ≠ "online advertising")
- **Semantic** = understands meaning & intent → finds "grow business online" even if document says "digital marketing strategies"

1. Preparation – done once

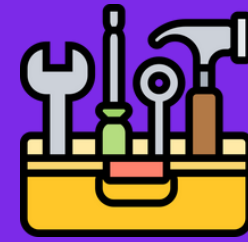


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2. When a user asks a question → RAG in action

- Convert question → embedding
- Search vector store → find most relevant chunks
- Give those chunks to the LLM as extra context

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→ The model no longer guesses — it answers using the actual relevant parts of your documents / manual.

RAG = external knowledge + LLM reasoning = more accurate, up-to-date, source-grounded answers